

Design Proposal for On-chip Unbalanced Mach-Zehnder Interferometers

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Abstract: We propose two sets of unbalanced Mach-Zehnder interferometers (MZIs) on the 220 nm silicon-on-insulator platform, which are built from 500-nm-wide TE strip waveguides. The path-length imbalances ΔL span from 19 to 297 μm , giving simulated free spectral ranges (FSR) from 30 nm down to 1.9 nm around 1550 nm. We calculated the waveguide compact model, and the MZI transfer function.

1. Introduction

A Mach-Zehnder Interferometer (MZI) in silicon photonics is an on-chip device that splits light into two waveguide arms and recombines it, so that the output intensity depends on the phase difference between the arms [1]. MZIs are fundamental building blocks of silicon photonic circuits, and are used in high-speed optical modulators [2] and on-chip photonic quantum computer [3]. When the two arms have different lengths, the MZI is called unbalanced, and its transmission becomes periodic in wavelength. The spacing of these fringes, the free spectral range (FSR), depends on the waveguide group index, which makes the unbalanced MZI a simple tool for measuring the waveguide dispersion.

In this report, we designed 6 different MZIs. All six devices are built from silicon strip waveguides with dimension of 500 nm $\times$ 220 nm and differ in their path-length imbalance ΔL , which ranges from about 19 μm to 297 μm . The design objective is to extract the waveguide group index from the free spectral range (FSR) of these unbalanced interferometers. Since silicon waveguide is dispersive, we also calculated the group index of the waveguides using Lumericals.

2. Theory

To model the MZI, we first consider an ideal Y-branch splitter and combiner. The guided light is treated as a plane wave propagating along the waveguide, $E(z, t) = E_0 \exp[i(\beta z - \omega t)]$, where $\beta = 2\pi n_{\text{eff}}/\lambda$ is the propagation constant of the waveguide mode. The transverse mode profile, propagation loss, and reflections are neglected. Let the input electric field be E_{in} . An ideal Y-branch splits it equally into the two arms, so

$$E_1 = E_2 = \frac{E_{\text{in}}}{\sqrt{2}}$$

The two beams travel through arms of length L_1 and L_2 and acquire phases βL_1 and βL_2 . The combiner recombines them with another factor of $1/\sqrt{2}$, so the output electric field is

$$E_{\text{out}} = \frac{1}{\sqrt{2}}(E_1 e^{-i\beta L_1} + E_2 e^{-i\beta L_2}) = \frac{E_{\text{in}}}{2}(e^{-i\beta L_1} + e^{-i\beta L_2})$$

The output intensity is $I_{\text{out}} = |E_{\text{out}}|^2$, which gives

$$I_{\text{out}} = I_{\text{in}} \cos^2\left(\frac{\beta \Delta L}{2}\right) = \frac{I_{\text{in}}}{2} \left[1 + \cos\left(\frac{2\pi n_{\text{eff}} \Delta L}{\lambda}\right)\right]$$

where $I_{\text{in}} = |E_{\text{in}}|^2$ is the input intensity, n_{eff} is the effective index of the waveguide, and $\Delta L = L_2 - L_1$ is the path-length difference between the two arms of the MZI. Therefore, the output

signal of an MZI depends on n_{eff} , ΔL , and the wavelength λ . We define the free spectral range (FSR) as [1]:

$$FSR(\lambda) = \frac{\lambda^2}{\Delta L \cdot n_g(\lambda)}$$

where λ is the wavelength and $n_g(\lambda)$ is the group index of the waveguide. The FSR is inversely proportional to ΔL , so a longer arm imbalance gives more closely spaced fringes.

3. Simulation

We choose the silicon strip waveguide on a buried oxide layer with SiO₂ cladding for our waveguide parameters. Its core is 500 nm wide and 220 nm high. We solved for the guided modes at $\lambda = 1550$ nm using the finite-difference eigenmode solver in Lumerical MODE Solutions. The calculated electric field profile of the transverse electric (TE) mode and transverse magnetic (TM) mode of the waveguide are shown in Fig. 1, and their corresponding effective indices are listed in Table 1.

Then, we swept the wavelength from 1500 to 1600 nm, and obtained the effective index and group index of TE mode of the waveguide (Fig. 2) for dispersion of the material and waveguide. By fitting the data with polynomial function, we obtained the following expression for compact model at $\lambda = 1550$ nm (the result is shown in Fig. 3):

$$n_{\text{eff}}(\lambda) = 2.4469 - 1.0363(\lambda - 1550\text{nm}) - 0.1379(\lambda - 1550\text{nm})^2$$

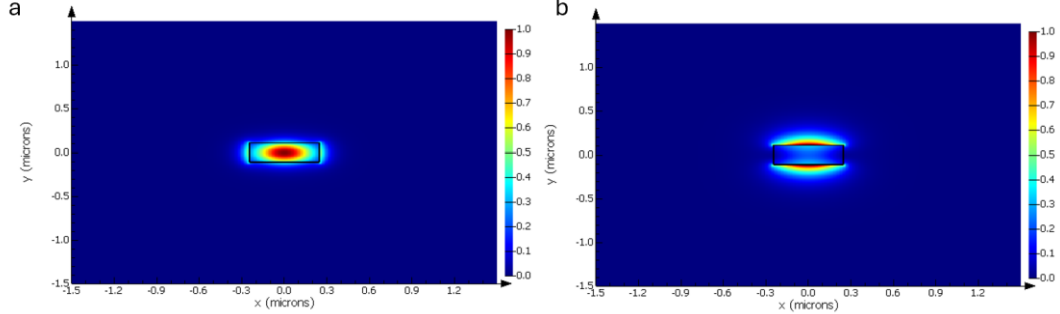


Fig. 1 The electric field profiles of the strip waveguide (220nmx500nm). (a) TE mode of the waveguide. (b) TM mode of the waveguide.

Mode	Effective index
TE1	2.4464
TM1	1.7761

Table 1 The calculated modes and their effective indices.

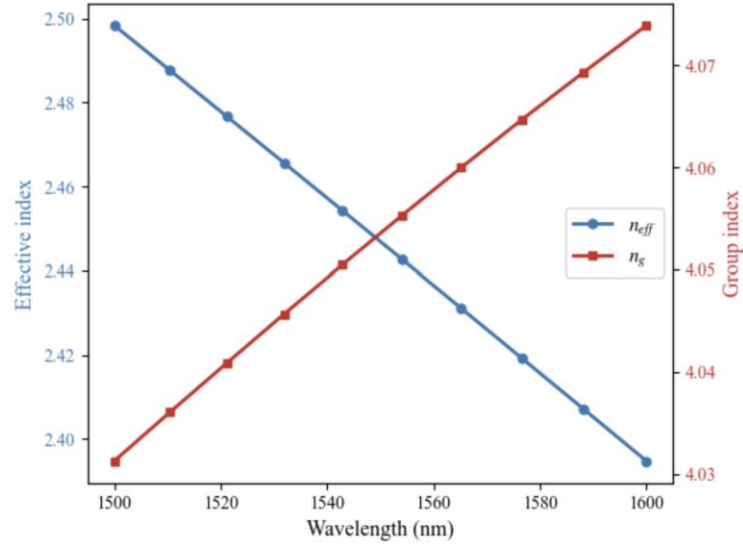


Fig. 2 Calculated effective index (blue) and group index (red) of TE waveguide mode as function of wavelength (nm)

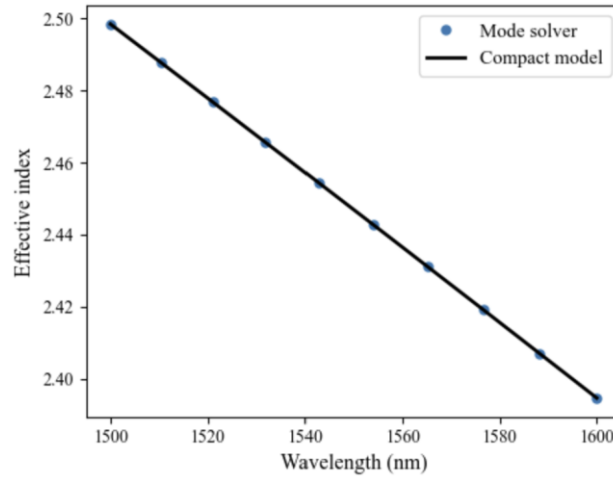


Fig.3 Effective index of the waveguide TE mode and the compact model.

4. Mach-Zehnder Interferometers

The KLayout of the devices are shown in Fig. 4 ($605 \times 410 \mu\text{m}$ -class chip area). Each device is fed by grating couplers (GC). MZI1, MZI2, and MZI3 (left and centre in Fig.) use a Y-branch splitter and a 2×2 combiner with an output on each side (one input grating coupler and two outputs). Next, MZI4, MZI5 and MZI6 (right) use 2×2 couplers at both ends with the unused port terminated. The imbalance is set by the long loops in one arm. Six variations are included, with ΔL chosen to give FSRs well below the 50 nm bandwidth (Table 2).

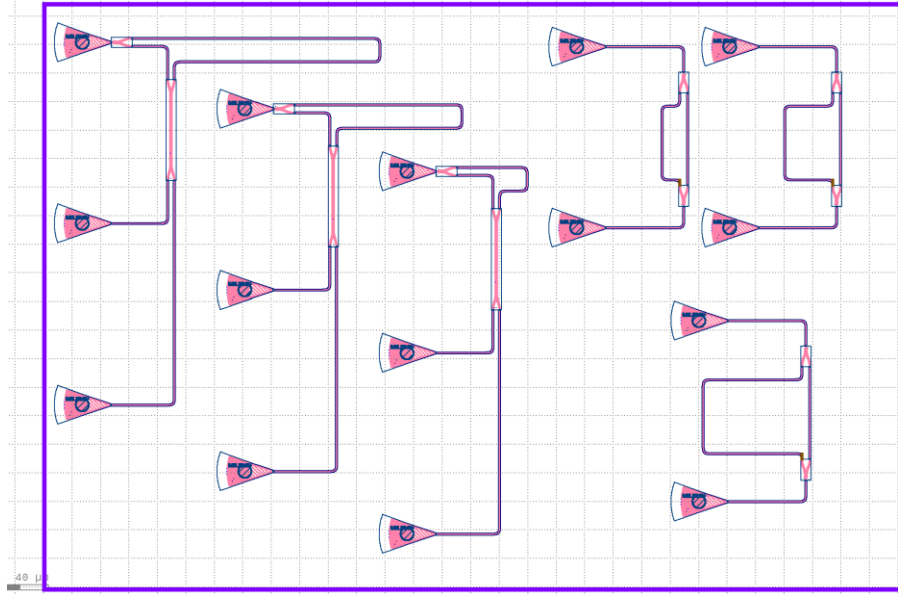


Fig. 4 The Klayout of the MZI designs

Device	ΔL (μm)	FSR (nm)
MZI1	297	1.94
MZI2	182	3.17
MZI3	45	12.85
MZI4	19	30.17
MZI5	62	9.31
MZI6	133	4.31

Table 2 The designed devices with their corresponding ΔL and FSR.

We performed a series of simulations for the designed devices using Lumerical INTERCONNECT, and the gain of the MZIs are shown in Fig. 5. Since FSR is inversely proportional to ΔL , we can clearly observe that the FSR is shorter as length mismatch is larger. For reference, we have also simulated the loss of the grating coupler, and the result is shown in Fig. 6.

5. Conclusion

We designed six unbalanced MZIs on the 220 nm silicon-on-insulator platform, using 500 nm wide TE strip waveguides with path-length differences from 19 to 297 μm . We also discuss the relation between free spectral range and path-length difference. The fabricated devices will be measured to find out which value describes the real waveguide, and the same extraction method will be used on the measured spectra.

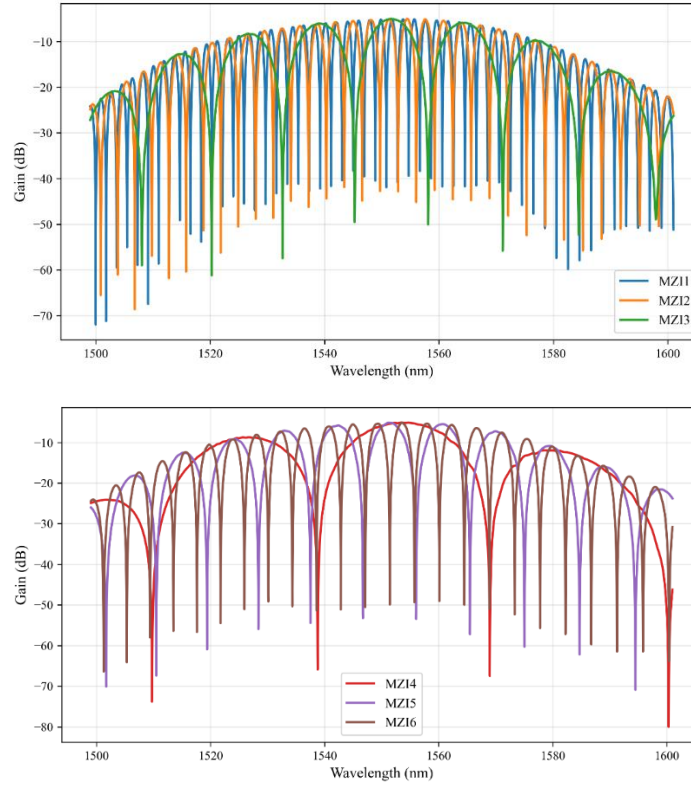


Fig. 5 The gain of the Mach-Zehner interferometers with different designs and ΔL .

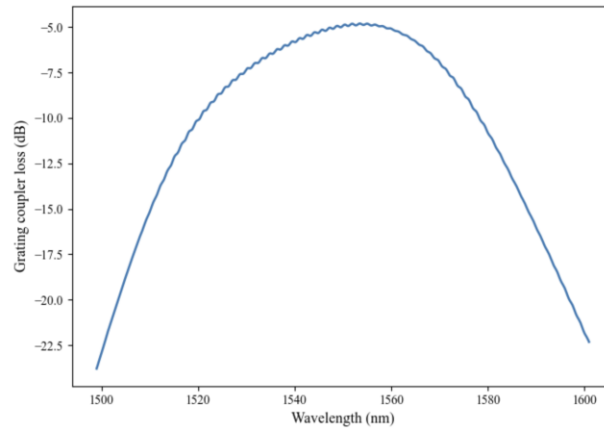


Fig. 6 The insertion loss from grating coupler.

References

- [1] Chrostowski, Lukas, and Michael Hochberg. *Silicon photonics design: from devices to systems*. Cambridge University Press, 2015.
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